

NBA/NHL Expected Points and Points Over Expected Analysis

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1 Abstract

This proposal evaluates shooting efficiency in the NBA and NHL with expected probability models and residual-based performance metrics. We use NBA xFG (logistic regression) and NHL xG (LASSO + GAM diagnostics) to estimate scoring probability at the shot level, then quantify over- and under-performance through Points/Goals Over Expected and Shot Difficulty Index (SDI). The current draft now includes evidence from both leagues: two NBA figures from the existing pipeline and two NHL figures generated from teammate data, each paired with concise interpretation. Deliverables are a reproducible R Markdown workflow, balanced cross-league analysis, and a submission-ready 2-3 page proposal PDF.

2 Introduction

Expected-value modeling lets us separate shot quality from shot outcomes. For the NBA, we analyze optical-tracking shot data from 2014-15 through 2025-26; for the NHL, we analyze MoneyPuck shot data from 2007-2024. This framing supports fair player comparison across roles, volume, and shot environments.

Table 1: Sample NBA Player Summary (2025-26)

Player	Shots	FG%	Total POE	POE/100
AJ Green	391	0.4398977	73.166894	18.7127606
Aaron Gordon	273	0.5091575	14.497316	5.3103721
Aaron Nesmith	370	0.3891892	-30.114935	-8.1391715
Aaron Wiggins	363	0.4435262	-3.078856	-0.8481698
Ace Bailey	518	0.4498069	-42.170340	-8.1409924
Ajay Mitchell	456	0.4868421	22.561261	4.9476449

3 Methods

We model NBA xFG with logistic regression and NHL xG with LASSO using shot context features (distance, angle, shot context, and game-state proxies). NHL shot-level `xGoal` probabilities are used directly in residual comparisons against observed goals. We then compute residual-style metrics to measure actual scoring above or below expectation at player level, and apply GAM diagnostics to validate non-linear spatial effects. For interpretability, both pipelines use SDI-style framing so efficiency can be evaluated against shot difficulty rather than raw conversion rate alone.

For the NHL visual analysis in this proposal, we read local shot-level data through a centralized resolver (NHL_CSV_PATH environment variable first, fallback paths second) and fail fast with a clear error if data is unavailable. This keeps the document portable for teammates while avoiding hardcoded machine paths or in-document downloads.

4 Expected Outcomes

4.1 NBA Visual Evidence

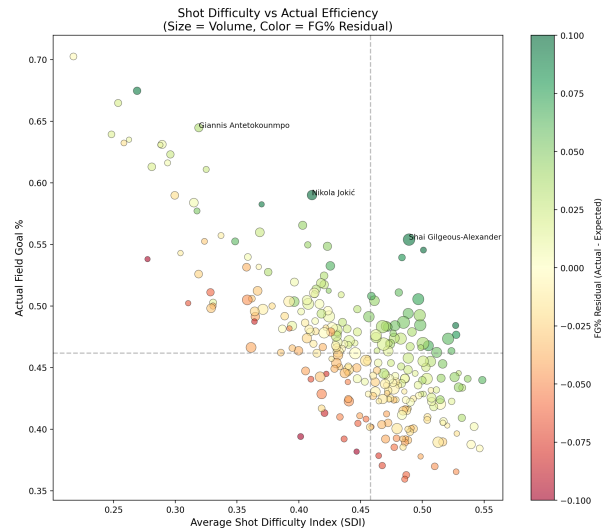


Figure 1: Figure 1: NBA Shot Difficulty Index vs Expected FG% (2025-26)

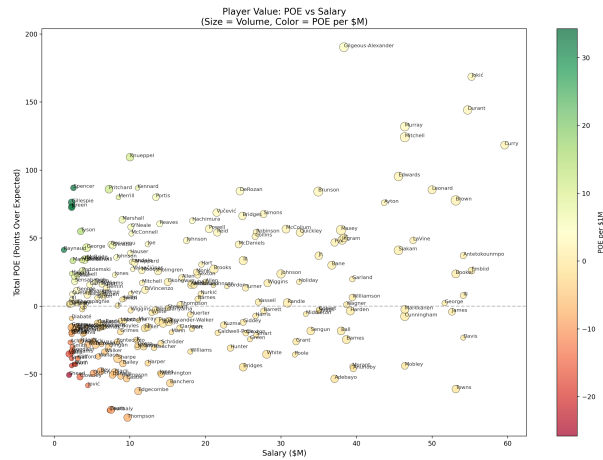


Figure 2: Figure 2: NBA Points Over Expected vs Salary (2025-26)

Figure 1 shows the expected inverse relationship between SDI and xFG%, confirming that shot context drives modeled efficiency in the NBA sample. Figure 2 highlights value dispersion at similar salary levels, motivating POE per salary context rather than salary-only evaluation. Positive outliers indicate players returning more scoring value than their contract tier suggests.

4.2 NHL Visual Evidence

Figure 3 mirrors the NBA pattern: players taking harder shots trend toward lower conversion rates. Figure 4 shows that higher NHL shot volume does not guarantee higher finishing efficiency. The weak linear trend supports evaluating players on shot quality and residual performance, not volume alone.

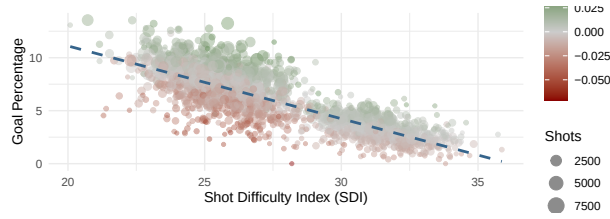


Figure 3: Figure 3: NHL Shot Difficulty Index vs Goal Percentage (players with ≥ 100 shots)

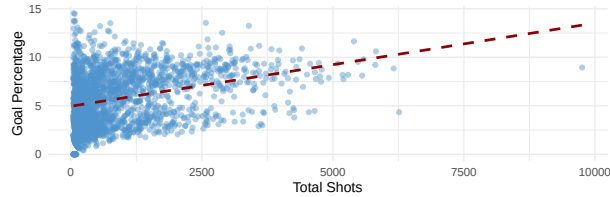


Figure 4: Figure 4: NHL Shot Volume vs Goal Percentage (players with ≥ 50 shots)

4.3 Cross-League Visual Summary

NBA Figure 1 and NHL Figure 3 both show the same directional pattern: higher shot difficulty is associated with lower average scoring probability. NBA Figure 2 and NHL Figure 4 then show why outcome interpretation must include value context and residuals, since volume and salary alone do not explain efficiency outliers. Together, these visuals justify a shared expected-points framework across both leagues.

5 Group Considerations

Our three-person team split work across data access, modeling, and communication deliverables so each league could be developed in parallel and then integrated in one document.

Task	Kanaan	Kaleb	Drianna
Data Collection (NBA/NHL)	0%	50%	50%
Feature Engineering + SDI	0%	50%	50%
Modeling (xFG/xG, diagnostics)	30%	40%	30%
Figures + Proposal Writing	33%	33%	33%

The immediate finalization step is validation of render quality, section completeness, and branch hygiene for merge into `main`.